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Maximum likelihood estimation of length-biased censored data under Frank copula dependency

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Abstract:

Ignoring censoring causes bias in survival analysis, especially for length-biased data. In this work, we estimate parameters in length-biased survival data with dependent censoring using a copula function to model the dependency between survival and censoring times. Extensive simulations show that our approach improves estimation accuracy and simplifies computing conditional probabilities of copulas. A real-data application is also provided.

Keywords: Survival analysis, Length-biased, Copula, Parameter estimation

Mathematics Subject Classification (2010): 62H05, 62Hxx, 62N02.

1 Introduction

Survival analysis is the analysis of time-to-event data such as the length of time from a time origin to an endpoint of interest. Such analytical methods are usually used to analyze data collected prospectively in time, such as data from a prospective cohort study or data collected for a clinical trial [Kartsonaki \(2016\)](#).

One of the main statistical challenges in survival analysis is handling censored data. This issue occurs when we have some information about individual survival time, but in some cases, we do not know the survival

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time exactly [Singh and Mukhopadhyay \(2011\)](#)

The main aim of survival analysis is focus to the time between an initiating event and a specific or terminating event, such as the time from the onset of a disease to relapse or death due to the disease, however, sometimes the onset is not clearly specified [Akbari et al. \(2025\)](#). In recent years, copula function was applied in survival analysis, especially on length biased. For some instances, [Rabhi and Bouezmarni \(2020\)](#) have used copulas for modeling dependence when the collected data are length-biased and account for both informative censoring.

The main aim of this paper is introduce statistical method for analyzing length biased data subject to dependent censoring. We express parametric likelihood method by assuming a copula between residual life and residual censoring time and assuming that left truncation time is independent of residual life and censoring time.

2 Distribution of length-biased survival data

Inspiring the the seminal work of [Sklar \(1959\)](#), for two random variables X and Y , there exists a function $C : [0, 1]^2 \rightarrow [0, 1]$ such that

$$F_{X,Y}(x, y) = C(F_X(x), F_Y(y)), \quad (2.1)$$

where $F_{X,Y} : \mathbb{R}^2 \rightarrow [0, 1]$ is the joint distribution function of the random vector (X, Y) and $F_X, F_Y : \mathbb{R} \rightarrow [0, 1]$ are distribution functions of random variables X and Y , respectively. The function $C : [0, 1]^2 \rightarrow [0, 1]$ is a copula if and only if it is a grounded, uniformly marginal and 2-increasing bivariate function. Hence their joint density copula, if exists, is just function $c(., .)$ such that

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)c(F_X(x), F_Y(y)). \quad (2.2)$$

Let $\tilde{T}$ be survival time in the unbiased population with probability density function $\tilde{f}(t)$, cumulative distribution $\tilde{F}(t)$, and survival function $\tilde{S}(t) = 1 - \tilde{F}(t)$. Suppose a random sampling time occurs uniformly over the calendar time axis. Let A denote the observed *truncation time* and V is the residual life and C is the residual censoring time after sampling. So, $T = A + V$ is the total survival time and $U = A + C$ is the total censoring time. In a random sample of n independent subjects, the length biased data in the absence of covariates consist of (y_i, A_i, δ_i) where $y_i = A_i + \min(V, C)$ and $\delta_i = I(V_i \leq C_i)$ is censoring indicator. This case is referred to as length-biased sampling. Biased sampling occurs when an investigator naturally collects samples from a population, but the sampling distribution is different from the target population. It happens because not every unit in the population has an equal chance to be sampled when the natural sampling plan is adopted [Qin \(2017\)](#). Because individuals with longer lifetimes are more likely to be present

at the sampling time, the observed distribution of T is known as length-biased. That is,

$$f_T^{(\text{obs})}(t) = \frac{t \tilde{f}(t)}{\mu}, \quad \mu = E[\tilde{T}] = \int_0^\infty t \tilde{f}(t) dt, \quad t > 0.$$

This is the standard form of the length-biased density and the conditional density of A given $T = t$ can be readily obtained as $f_{A|T}(a | t) = \frac{1}{t}$, $0 < a < t$, which means the sampling time is equally likely to fall anywhere in the interval $(0, t)$. Also, by the aid of multiplication rule, their joint density is obtained as

$$f_{(A,T)}(a, t) = \frac{\tilde{f}(t)}{\mu}, \quad t > a > 0,$$

Integrating over t of this density helps us to acquire the marginal density of A as $f_A(a) = \frac{\tilde{S}(a)}{\mu}$, $a > 0$. Similarly, since $V = T - A$, the same expression holds for $f_V(v)$ can be represented as $f_V(v) = \frac{\tilde{S}(v)}{\mu}$, $v > 0$. Thus we have

$$f_A(t | X) = f_V(t | X) = \frac{\tilde{S}(t | X)}{\mu(X)}, \quad t > 0,$$

We use these densities to explore the parameter estimation in length-biased censored survival data using a copula-based approach. For this, we follow [Emura and Chen \(2018\)](#) and consider a bivariate survival function

$$P(V > v, C > c) = C_\theta(S_V(v), S_C(c)), \quad (2.3)$$

where $S_V(v) = P(V > v)$, $S_C(c) = P(C > c)$ are respectively the marginal survival function of V and C , and C_θ is a copula with parameter θ . According to (2.3) and using facts $T = A + V$ and $U = A + C$, via some algebra we obtain the joint survival of T and U using the copula of V and C as

$$P(T > t, U > t) = \int C_\theta(S_V(t-a), S_C(t-a)) f_A(a) da. \quad (2.4)$$

We apply these relations to carry out a parameter estimation. For n independent cases and their corresponding observed values (y_i, A_i, δ_i) , $i = 1, 2, \dots, n$, the likelihood function is constructed. The likelihood can be used to estimate parameters of the length-biased model.

The following proposition explores our results under the exponential marginals and Frank copula.

Proposition 2.1. *Let marginal distributions of random variables V and C follow exponential as $S_V(v, \lambda_1) = e^{-\lambda_1 v}$ and $S_C(c, \lambda_2) = e^{-\lambda_2 c}$ where $\lambda_1 > 0$ and $\lambda_2 > 0$ are their parameters respectively. Also assume that they are connected via the Frank copula. Then log-likelihood function for observed values (y_i, A_i, δ_i) , $i = 1, 2, \dots, n$, is Defined as follows:*

$$\begin{aligned} \ell = \sum_{i=1}^n \delta_i \log \int \frac{\lambda_1 S_1^{(i)} e^{-\theta S_1^{(i)}} (e^{-\theta S_2^{(i)}} - 1)}{(e^{-\theta} - 1) + (e^{-\theta S_1^{(i)}} - 1) (e^{-\theta S_2^{(i)}} - 1)} f_A(a_i) da_i \\ + (1 - \delta_i) \log \int \frac{\lambda_2 S_2^{(i)} e^{-\theta S_2^{(i)}} (e^{-\theta S_1^{(i)}} - 1)}{(e^{-\theta} - 1) + (e^{-\theta S_1^{(i)}} - 1) (e^{-\theta S_2^{(i)}} - 1)} f_A(a_i) da_i, \end{aligned}$$

where $S_j^{(i)} = e^{-\lambda_j(y_i - a_i)}$, $j = 1, 2$ are the corresponding survival functions.

3 Numerical Analysis

In this section we provide the numerical studies, including simulated analysis and real data application.

3.1 Simulation Study

In order to explain and assess our approach, in this section we used the Mont-Carlo method in our simulation study. We considered exponential marginals and Frank copula. In this scenario, we supposed that residual life V and residual censoring C follow exponential distributions with parameters λ_1 and λ_2 respectively and truncation time A has uniform distribution on $(0.1, 10.1)$. We also assume that their dependency is modeled by the Frank copula with parameter θ . Assuming $\lambda_1 = 1.2$ and $\lambda_2 = 1.5$, we considered two scenarios for the dependency structure, one with a positive relationship and one with negative relationship; which we assumed $\theta = 5.5$ for relationship. We generated 1000 samples and by constructing the objective function (2.3) we estimated parameters. In order to estimate a stable and robust estimator, we repeated $M = 50, 100$ and 200 times our parameter estimation. Under the assumption of $\theta = 5.5$, Table 1 provides the mean of

Repetition									
Parameter	$M = 50$			$M = 100$			$M = 200$		
	θ	λ_1	λ_2	θ	λ_1	λ_2	θ	λ_1	λ_2
Estimate	5.492	1.200	1.499	5.503	1.211	1.514	5.505	1.206	1.505
Bias	-0.007	0.0002	-0.0006	0.003	0.011	0.014	0.005	0.006	0.005
MSE	0.014	0.007	0.012	0.007	0.005	0.007	0.009	0.006	0.008

Table 1: The estimated values of parameters and their biases and MSE.

our estimated parameters as well as their bias and mean square error (MSE) for $M = 50, 100, 200$, see also Figure 1.

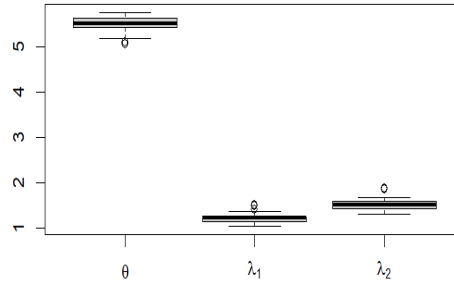


Figure 1: Convergence of the parameter estimation in length-biased simulated data with Frank copula and exponential marginals.

3.2 Real Data

To demonstrate the proposed method, we applied it to real data namely, Rotterdam data set, which is available in the `survival` package in R. This data set is a classic resource for studying breast cancer prognosis under left truncation and right censoring, and we used it to illustrate our approach. The data set has 2982 primary breast cancer patients treated at the Rotterdam Tumor Bank. We considered “rtime” as the survival time, “death” as censoring indicator and “year” as the left truncation time. For fitting marginal distributions of survival time (V) and censoring time (C), we chose four competitive distributions: Exponential, Weibull, Gamma and Lognormal. Based on the results of the top panel of Table 2, we realized that both survival and censoring times follow Weibull distributions. So, we proceed with parameter estimation by optimizing the log likelihood function by considering the Weibull distribution as the marginal. To model their dependency, in this optimization process we considered three copulas: Frank, Joe and Ali Mikhail Haq (AMH). The results are represented in the Table 2.

Results for the Rotterdam data set									
		Parameter estimates					Criteria		
Data set	Copula	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\theta}$	LL	AIC	BIC
Rotterdam	Frank	1.04	1.093	25.5	18.20	-20.35	20425.59	40865.18	40899.54
	Joe	1.03	1.089	27.18	19.033	1.588	145916.5	291847.1	291881.3
	AMH	1.039	1.089	26.95	18.998	-0.982	21067.53	42149.06	42183.42

Table 2: The ML estimated values of parameters and criteria for assuming different joint distributions for Rotterdam data set.

Discussion and Conclusions

We modeled the dependency of survival time and censoring time in length-biased data using a copula function. While we assumed truncation time is independent of residual time and censoring, this can be extended. Future work may consider the case where truncation time depends on residual time, residual censoring, or both.

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